Autoimmune Disease, Multisystem, Infantile-Onset, 3 (ADMIO3): A Comprehensive Disease Report

MONDO:0957388 · OMIM:620430 · Causal gene: *CBLB* (HGNC:1542)

Summary

Autoimmune disease, multisystem, infantile-onset, 3 (ADMIO3) is a rare, autosomal-recessive inborn error of immunity caused by **biallelic loss-of-function variants in *CBLB***, the gene on chromosome 3q13.11 encoding the E3 ubiquitin ligase **CBL-B (Casitas B-lineage lymphoma proto-oncogene-b)**. The disease name is a serial number and does not itself name the gene; per the identifiers supplied, the correct causal gene is ***CBLB***, and this report addresses that gene throughout. The disease was defined as a distinct Mendelian entity by Janssen and colleagues in 2022, who described three unrelated children with early-onset multisystem autoimmunity carrying distinct homozygous *CBLB* mutations (PMID: 36006710).

CBL-B is the central intracellular negative checkpoint of antigen-receptor signaling. By catalyzing ubiquitination of proximal signaling components (TCR-ζ, Vav1, PI3K, PKCθ/PLCγ1, CARMA1) it raises the activation threshold of lymphocytes, uncouples T-cell activation from the normal CD28 costimulation requirement, enforces T-cell anergy, and maintains sensitivity of effector T cells to regulatory-T-cell (Treg) suppression. When CBL-B catalytic (E3 ligase) function is lost on both alleles, this brake is removed: T cells hyperproliferate, resist Treg suppression, escape anergy, and B cells, NK cells and mast cells become hyperactivated — producing infantile-onset multisystem autoimmunity, sometimes with allergic features such as chronic urticaria and a predisposition to anaphylaxis.

The mechanism is strongly corroborated across species and model systems. The *Cblb*-knockout mouse spontaneously develops multi-organ autoimmunity with autoantibodies, lymphocytic tissue infiltration, IL-2 hyperproduction and high susceptibility to experimental autoimmune encephalomyelitis (EAE); a knock-in mouse bearing the orthologue of the patient p.H285L allele reproduces the human cellular phenotype; and the naturally occurring *Cblb*-nonsense-

mutant Komeda diabetes-prone (KDP) rat develops autoimmune diabetes, thyroiditis and alopecia. Common hypomorphic *CBLB* variants are, in turn, established susceptibility alleles for polygenic autoimmune diseases (multiple sclerosis, type 1 diabetes, systemic lupus erythematosus), placing ADMIO3 at the severe, monogenic pole of a CBL-B "tolerance rheostat" whose opposite pole is deliberate pharmacologic CBL-B inhibition to boost anti-tumour immunity.

Key Findings

Finding 1 — ADMIO3 is caused by biallelic loss-of-function *CBLB* variants and presents as infantile-onset multisystem autoimmunity

The gene-defining study (Janssen et al., 2022) identified distinct homozygous *CBLB* mutations in **3 unrelated children** with early-onset autoimmunity, one of whom also had chronic urticaria. The paper states: *"Distinct homozygous mutations in CBLB were identified in 3 unrelated children with early-onset autoimmunity, one of whom also had chronic urticaria"* and concludes that *"This study establishes CBL-B deficiency as a cause of immune dysregulation."* The three index alleles were **p.R496X** (nonsense; abolishes CBL-B expression), **p.C464W** (RING-finger; protein preserved but catalytically non-functional) and **p.H285L** (SH2/TKB domain; protein expressed at approximately half-normal level). Patient T cells hyperproliferated in response to anti-CD3 cross-linking, consistent with loss of the inhibitory brake. Inheritance is autosomal recessive. This is the primary human evidence establishing ADMIO3 (PMID: 36006710).

Finding 2 — CBL-B is an E3 ubiquitin ligase acting as the central negative checkpoint of antigen-receptor signaling; its loss drives T-cell hyperactivation and failure of anergy

CBL-B was the first E3 ubiquitin ligase directly implicated in peripheral T-cell activation and tolerance. Selective inactivation of its catalytic E3 activity causes *"T cell hyperactivation, spontaneous autoimmunity, and impaired induction of T cell anergy in vivo"* (PMID: 21248250). CBL-B is expressed across all leukocyte subsets, and *"cblb-deficient immune cells display lower activation thresholds and cblb knockout mice spontaneously develop autoimmunity and are highly susceptible to experimental autoimmunity"* (PMID: 25815272). Mechanistically, CBL-B normally couples T-cell-receptor (TCR) signaling to the requirement for CD28 costimulation; its

loss *"frees antigen receptor-triggered receptor clustering, lipid raft aggregation, and sustained tyrosine phosphorylation from the requirement for CD28 costimulation"* (PMID: 11070165). At the substrate level, CBL-B mediates **K33-linked polyubiquitination of TCR-ζ**, which *"affected its phosphorylation and association with the zeta chain-associated protein kinase Zap-70"* — a proteolysis-independent signaling control (PMID: 20637659).

Finding 3 — Patient effector T cells and a matched knock-in mouse show Treg-resistance and mast-cell hyperactivation, explaining autoimmunity plus urticaria/anaphylaxis

In Janssen 2022, a **CblbH257L knock-in mouse** (orthologous to the patient p.H285L allele) showed T- and B-cell hyperproliferation to antigen-receptor cross-linking. Although Tregs were *increased* in percentage and retained normal in-vitro suppressive function, the effector T cells were intrinsically resistant to suppression: *"T effector cells from the patient with the p.H285L mutation and CblbH257L mice were resistant to suppression by WT Tregs."* The same model linked CBL-B loss to allergic disease: *"Bone marrow-derived mast cells from CblbH257L mice were hyperactivated after FcεRI cross-linking, and CblbH257L mice demonstrated exaggerated IgE-mediated passive anaphylaxis"* (PMID: 36006710). Independent work confirms the Teff-intrinsic defect: *"Cbl-b(-/-) Teffs are resistant to suppression by Tregs in vivo"* (PMID: 20624942). This dual mechanism — Treg-resistant effector T cells plus hyperactive mast cells — accounts for both the autoimmune and the allergic (urticaria/anaphylaxis) components of the clinical picture.

Finding 4 — Common *CBLB* variants are established autoimmunity susceptibility alleles, situating the Mendelian disease on an allelic series

A Sardinian genome-wide association study (GWAS) linked the common *CBLB* variant **rs9657904** to multiple sclerosis (MS) at genome-wide significance (*"rs9657904, overall P = 1.60 x 10(-10), OR = 1.40"*) (PMID: 20453840). This was replicated in the Italian mainland (rs9657904 OR 1.31, 95% CI 1.14–1.52; PMID: 21037273) and in a pooled Spanish cohort (rs9657904 OR 0.89; PMID: 22194214). Mechanistically, the MS-risk SNP **rs12487066** lowers CBL-B expression: *"The MS risk-related single nucleotide polymorphism of CBLB rs12487066 is associated with diminished CBL-B expression levels and alters the effects of type I IFNs on human CD4(+) T cell proliferation"* (PMID: 25261476). Thus common hypomorphic variants cause polygenic autoimmune risk, while biallelic null variants cause the monogenic ADMIO3 — one continuous allelic series.

Finding 5 — *Cblb*-null mice recapitulate the human disease: spontaneous multi-organ autoimmunity

Two landmark 2000 *Nature* papers established the mouse model. Bachmaier et al. showed that *"gene-targeted mice lacking the adaptor Cbl-b develop spontaneous autoimmunity characterized by auto-antibody production, infiltration of activated T and B lymphocytes into multiple organs, and parenchymal damage,"* and that *"Mutation of Cbl-b uncouples T-cell proliferation, interleukin-2 production and phosphorylation of the GDP/GTP exchange factor Vav1 from the requirement for CD28 co-stimulation"* (PMID: 10646608). Chiang et al. showed CBL-B-deficient T cells no longer need CD28 for IL-2 production, that Vav activation is enhanced, and that *"Mice deficient in Cbl-b are highly susceptible to experimental autoimmune encephalomyelitis, suggesting that the dysregulation of signalling pathways modulated by Cbl-b may also contribute to human autoimmune diseases such as multiple sclerosis"* (PMID: 10646609).

Finding 6 — CBL-B is a multi-lineage immune checkpoint with substrates across T, NK, NKT, B and myeloid cells

CBL-B's regulatory reach extends well beyond the TCR. Confirmed substrates and nodes include TCR-ζ (K33 polyubiquitination; PMID: 20637659), Vav1 (PMID: 10646608), PKCθ and PLCγ1 (PMID: 25000980), CARMA1 (*"Cbl-b binds and promotes monoubiquitination to CARMA1, a critical signaling molecule in NFkappaB activation"* in NKT anergy; PMID: 19815501), and the TAM receptors in NK cells (*"The TAM tyrosine kinase receptors Tyro3, Axl and Mer... were identified as ubiquitylation substrates for Cbl-b"*; PMID: 24553136). Because *"Cbl-b is expressed in all leukocyte subsets and regulates several signaling pathways in T cells, NK cells, B cells, and different types of myeloid cells"* (PMID: 25815272), its loss produces genuinely multisystem disease, not just a T-cell disorder.

Finding 7 — The Komeda diabetes-prone (KDP) rat is a naturally occurring *Cblb*-nonsense-mutant disease model

Positional cloning identified a nonsense mutation in the rat *Cblb* gene (Iddm/kdp1) as one of two major susceptibility genes (with the MHC RT1u haplotype) for autoimmune type 1 diabetes in the KDP rat (PMID: 15897618). Congenic transfer reconstitutes insulitis and thyroiditis (PMID: 17259398); on a PVG.R23 background, *"The congenic rats with homozygous Cblb mutation showed autoimmune thyroiditis without T1D and slight to severe alopecia"* (PMID:

21918539). Human *CBLB* is likewise a T1D susceptibility locus: *"One SNP in exon 12 of the CBLB gene was significantly demonstrated to be associated to T1D in a large Danish T1D family material of 480 families"* (PMID: 15629882).

Finding 8 — Patient mutations map onto CBL-B's functional domains, all converging on loss of E3 ligase function

Cbl-family proteins share an N-terminal tyrosine-kinase-binding (TKB/phosphotyrosine-binding) domain, a linker helix, a zinc-coordinating RING-finger domain that carries E3 activity, and proline-rich/leucine-zipper regions; CBL-B *"retains the conserved phosphotyrosine binding domain (PTB) in the N-terminal and the zinc finger"* (PMID: 10362357). E3 activity is autoinhibited by the TKB and linker-helix domains and relieved by tyrosine phosphorylation: *"the E3 activity of c-Cbl is negatively regulated by other domains present in the amino-terminal half of the protein (the TKB and linker helix domains) and that this negative regulation is removed when the protein is phosphorylated on tyrosine residues"* (PMID: 15117950). The three ADMIO3 alleles thus hit three different structural modules — p.H285L (SH2/TKB), p.C464W (RING), p.R496X (truncation) — but all converge on **loss of catalytic E3 ligase function**.

Finding 9 — ADMIO3 is the loss-of-function mirror image of a validated druggable immune checkpoint

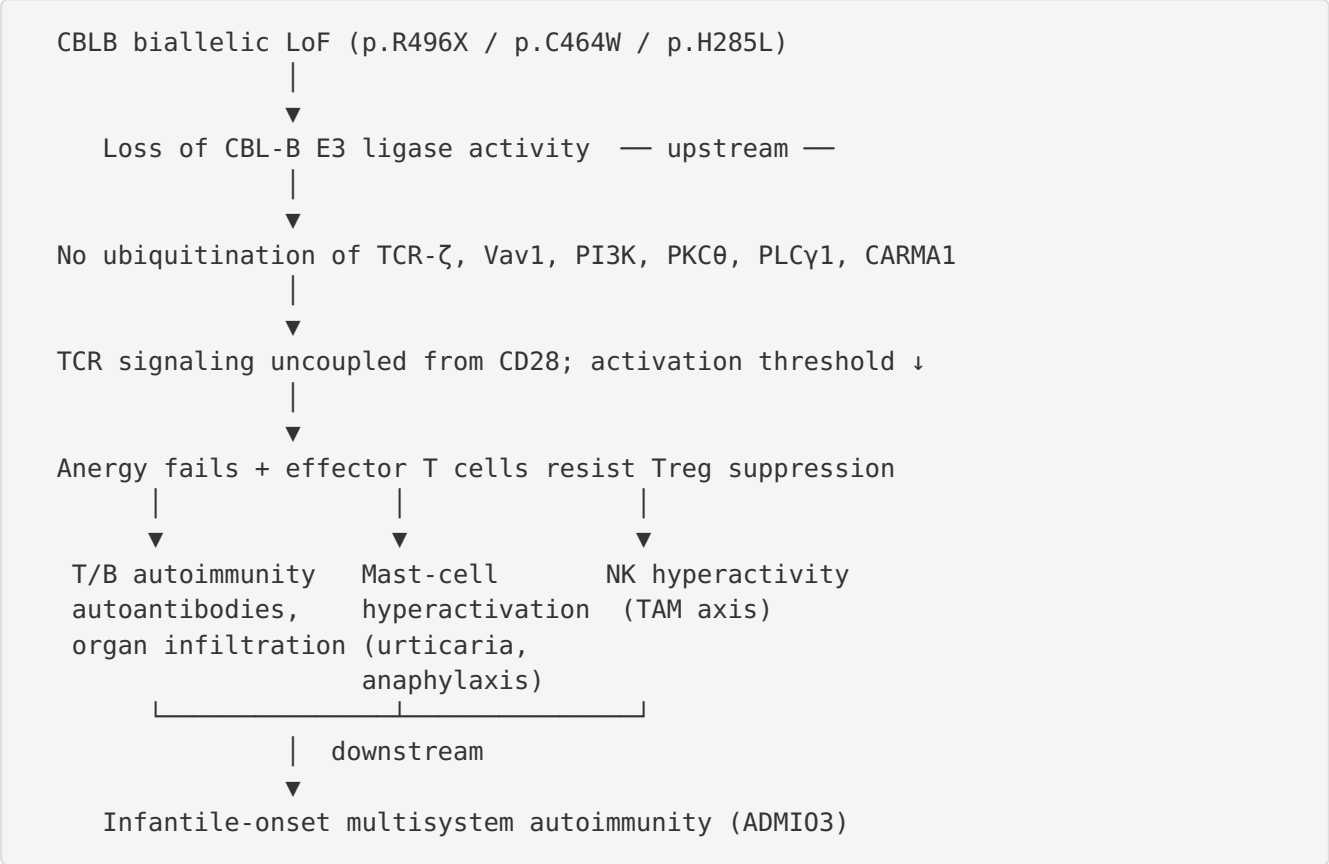
CBL-B loss-of-function in humans causes infantile multisystem autoimmunity, whereas *pharmacologic or genetic inactivation* of CBL-B is being developed to **enhance** anti-tumour immunity: *"genetic deletion of the E3 ubiquitin ligase Cbl-b... or targeted inactivation of its E3 ligase activity licenses natural killer (NK) cells to spontaneously reject metastatic tumours"* (PMID: 24553136), and *"the increased activation potential of cblb-deficient cells renders them more potent to fight against malignancies or infections"* (PMID: 25815272). ADMIO3 therefore sits at one extreme of a dose/function rheostat, which has direct therapeutic implications (see §Treatment).

Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function mutation in *CBLB*** (nonsense p.R496X, RING-finger p.C464W, or TKB p.H285L) → **loss or catalytic inactivation of the CBL-B E3 ubiquitin ligase** in lymphocytes and myeloid cells. *(demonstrated — human; PMID: 36006710)*
2. Loss of CBL-B catalytic activity → **failure to ubiquitinate proximal antigen-receptor signaling substrates** (TCR-ζ via K33 chains, Vav1, PI3K, PKCθ, PLCγ1, CARMA1). *(demonstrated — in vitro/mouse; PMID: 20637659, PMID: 25000980, PMID: 19815501)*
3. Failure of substrate ubiquitination → **TCR signaling is uncoupled from the CD28 costimulation requirement; receptor clustering, lipid-raft aggregation and sustained tyrosine phosphorylation proceed unchecked** → lowered activation threshold. *(demonstrated — mouse; PMID: 11070165, PMID: 10646608)*
4. Lowered activation threshold → **impaired induction/maintenance of T-cell anergy and effector T cells become resistant to Treg-mediated suppression** (despite numerically normal/increased Tregs). *(demonstrated — human + mouse; PMID: 21248250, PMID: 36006710, PMID: 20624942)*
5. **Branch A (T/B cells):** Treg-resistant, anergy-escaping lymphocytes hyperproliferate, hyperproduce IL-2, generate autoantibodies and infiltrate multiple organs → **multisystem autoimmunity**. *(demonstrated — mouse; PMID: 10646608)*
6. **Branch B (mast cells/myeloid):** CBL-B loss hyperactivates mast cells after FcεRI cross-linking → **chronic urticaria and exaggerated IgE-mediated anaphylaxis**. *(demonstrated — mouse model of patient allele; PMID: 36006710)*
7. **Branch C (NK cells):** loss of TAM-receptor (Tyro3/Axl/Mer) ubiquitination raises NK activation — the same axis exploited therapeutically in cancer — contributing to immune dysregulation. *(inferred for human disease; demonstrated mechanism in mouse; PMID: 24553136)*
8. Convergent organ damage → **infantile-onset clinical autoimmune/inflammatory disease** (ADMIO3). *(demonstrated — human; PMID: 36006710)*

Schematic



Upstream vs downstream

- **Upstream (initiating):** *CBLB* mutation → loss of E3 ligase activity → failed substrate ubiquitination.
- **Midstream (amplifying):** lowered activation threshold, CD28-independence, anergy failure, Treg resistance.
- **Downstream (effector/clinical):** autoantibody production, multi-organ lymphocytic infiltration, mast-cell-driven allergy, tissue damage.

Suggested ontology terms

- **Gene/protein:** *CBLB* (HGNC:1542); CBL-B E3 ubiquitin-protein ligase (UniProt Q13191).
- **GO biological process:** protein polyubiquitination (GO:0000209); negative regulation of T cell activation (GO:0050868); T cell anergy (GO:0002870); regulation of immune response (GO:0050776); negative regulation of T cell receptor signaling pathway (GO:0050860).
- **GO cellular component:** cytosol (GO:0005829); plasma membrane (GO:0005886).

- **CL cell types:** T cell (CL:0000084), CD4-positive alpha-beta T cell (CL:0000624), regulatory T cell (CL:0000815), B cell (CL:0000236), natural killer cell (CL:0000623), mast cell (CL:0000097), dendritic cell (CL:0000451).
- **Disease:** MONDO:0957388; autoimmune disease (MONDO:0005952, parent).

Anatomical Structures, Phenotypes and Temporal Profile

Phenotypes (from the defining cohort and model systems). The disease is a **multisystem** autoimmune/immune-dysregulation syndrome with **infantile onset** (HPO: Infantile onset, HP:0003593; Autoimmunity, HP:0002960). Reported/model-supported features include: early-onset autoimmunity (HP:0002960), chronic urticaria (HP:0001078) in at least one patient, and — by extrapolation from the *Cblb*-mutant rat and knockout mouse — organ-specific autoimmune targets such as autoimmune thyroiditis (HP:0100646), type 1 diabetes/insulitis (HP:0100651), and alopecia (HP:0001596). Laboratory hallmarks include autoantibody production (HP:0030057), lymphocyte hyperproliferation to antigen-receptor stimulation, and (in the matched mouse model) exaggerated IgE-mediated anaphylaxis and elevated Tregs with intact intrinsic suppressive function but Treg-resistant effector T cells. Because only three index patients are published, precise per-phenotype frequencies, severity grading and quality-of-life data are **not yet available** for the human disease.

Anatomy / organ systems (UBERON). Primary target is the **immune system** (UBERON:0002405) — peripheral lymphoid organs including spleen (UBERON:0002106) and lymph nodes (UBERON:0000029). Secondary organ involvement is inferred from model data and the multisystem designation: thyroid gland (UBERON:0002046), pancreatic islets (UBERON:0000006), skin (UBERON:0002097; urticaria, alopecia), and potentially central nervous system/white matter in the MS-analogous EAE susceptibility seen in mice (UBERON:0002316). **Subcellular:** CBL-B acts in the cytosol and at the inner plasma membrane of leukocytes; its substrates reside in the proximal antigen-receptor signalosome.

Temporal development. Onset is **infantile/early childhood** by definition. The limited human data describe a chronic course; model data suggest a progressive, multi-organ autoimmune evolution rather than a self-limited one. Exact progression rate, relapsing-remitting vs. progressive pattern, remission behaviour and critical intervention windows are **not established** in humans given the cohort size.

Genetics, Inheritance and Population

- **Causal gene:** *CBLB* (HGNC:1542; NCBI Gene 868; Ensembl ENSG00000114423; chromosome 3q13.11), encoding the **E3 ubiquitin-protein ligase CBL-B** (UniProt Q13191; OMIM gene 604491). Disease OMIM **620430**.
- **Index pathogenic variants (all germline, homozygous, autosomal recessive):**

Variant (protein)	Type	Domain	Effect on CBL-B	Classification
p.R496X	Nonsense / truncating	Downstream of RING	Abolishes protein expression	Pathogenic (LoF)
p.C464W	Missense	RING finger (Zn-coordinating)	Protein preserved, E3 activity lost	Pathogenic (LoF)
p.H285L	Missense	SH2 / TKB (PTB)	~50% protein, hypomorphic	Pathogenic (LoF)

- **Functional consequence:** uniformly **loss of function** of the E3 ligase; no gain-of-function or dominant-negative mechanism described. (PMID: 36006710, PMID: 15117950)
- **Inheritance:** autosomal recessive; heterozygous carriers are healthy. Penetrance in biallelic carriers appears high but cannot be quantified from n = 3.
- **Allelic series / modifier context:** common hypomorphic regulatory variants (rs9657904, rs12487066, exon-12 SNP) are susceptibility alleles for MS, T1D and SLE (PMID: 20453840, PMID: 25261476, PMID: 15629882); human CBLB genetically interacts with CTLA4 in T1D (PMID: 15629882).
- **Epidemiology:** ADMIO3 is **ultra-rare** — only three published families to date. Prevalence/incidence, founder effects, carrier frequency, sex ratio and geographic distribution are **not established**. Consanguinity is a plausible contributor given homozygosity but was not systematically reported.

Diagnostics

- **Genetic testing is the definitive diagnostic modality:** whole-exome or whole-genome sequencing, or a comprehensive inborn-errors-of-immunity (IEI) gene panel that includes

CBLB, with confirmation of **biallelic** pathogenic variants. Single-gene *CBLB* sequencing is appropriate when the phenotype is highly suggestive or a familial variant is known.

- **Supportive immunologic workup:** autoantibody panels (autoimmunity screen), total/specific IgE (given urticaria/anaphylaxis risk), and functional assays showing **T-cell hyperproliferation to anti-CD3/anti-CD28 cross-linking** and **effector-T-cell resistance to Treg suppression** — the laboratory signatures demonstrated in patients and the matched mouse (PMID: 36006710).
- **Differential diagnosis:** other monogenic immune-dysregulation/IPEX-like disorders (FOXP3, CTLA4 haploinsufficiency, LRBA deficiency, STAT1/STAT3 GOF), and the other "autoimmune disease, multisystem, infantile-onset" entities (types 1 and 2, different genes). The *CBLB* genotype distinguishes ADMIO3.

Treatment and Prevention

No ADMIO3-specific approved therapy or clinical trial exists; management is extrapolated from immune-dysregulation principles and the mechanism.

- **Mechanism-guided rationale:** because the lesion removes the inhibitory brake and renders effector T cells **CD28-costimulation-independent** and **Treg-resistant**, rational therapy aims to **restore inhibitory tone**. Candidate approaches include **CTLA4-Ig / abatacept** (NCIT: Abatacept, C47384) to re-impose the costimulation brake, and **mTOR inhibition (sirolimus/rapamycin; NCIT: Sirolimus, C1212)** — supported indirectly by SLE studies where rapamycin plus vitamin D restored Treg-related anergy factors (PMID: 25351606). Standard immunosuppression (corticosteroids, calcineurin inhibitors) and allergy management (antihistamines, anti-IgE for mast-cell features) are reasonable symptomatic measures. (PMID: 36006710)
- **Therapeutic inversion / caution:** ADMIO3 is the **loss-of-function mirror** of oncology CBL-B **inhibitors** developed to boost anti-tumour T/NK immunity (PMID: 24553136, PMID: 25815272); CBL-B agonism/restoration is the conceptual therapeutic direction for the disease, and CBL-B inhibitors are strictly contraindicated.
- **Prevention/counseling:** autosomal-recessive genetic counseling, carrier testing of relatives, and prenatal/preimplantation options where a familial variant is known. No primary prevention beyond reproductive counseling is applicable to a monogenic disorder.

Model Organisms and Comparative Biology

Model	Lesion	Key phenotype	PMID
Cblb−/− mouse (knockout)	Full null	Spontaneous multi-organ autoimmunity, autoantibodies, IL-2 hyperproduction, CD28-independence, EAE susceptibility	10646608, 10646609
CblbH257L knock-in mouse	Orthologue of patient p.H285L	T/B hyperproliferation; Treg-resistant effector T cells; mast-cell hyperactivation; exaggerated IgE anaphylaxis	36006710
Catalytic-dead (E3-inactive) mouse	RING inactivation	T-cell hyperactivation, spontaneous autoimmunity, impaired anergy	21248250
Komeda diabetes-prone (KDP) rat	Natural *Cblb* nonsense	Autoimmune T1D, thyroiditis, alopecia (with MHC RT1u)	15897618, 21918539

Orthologues: mouse *Cblb* (NCBI Gene 208650), rat *Cblb* (NCBI Gene 171109). Disease mechanisms are **evolutionarily conserved** across human, mouse and rat, and the KDP rat represents a **naturally occurring** animal disease of veterinary/comparative interest (OMIA relevance). Phenotype recapitulation is strong for the core immune-dysregulation and multi-organ autoimmunity; model limitations include species differences in which organs are preferentially targeted and the small human dataset against which to benchmark.

Evidence Base

PMID	Title (abbrev.)	Evidence type	Role in this report
36006710	*Immune dysregulation caused by homozygous mutations in CBLB*	Human clinical + matched mouse	**Gene-defining** study; variants, Treg-resistance, mast-cell phenotype
21248250	*Essential role of E3 ligase activity in Cbl-b-regulated T cell functions*	Mouse	E3 activity required to prevent autoimmunity/anergy failure
11070165	*Cbl-b is a negative regulator of receptor clustering...*	Mouse/in vitro	CD28-independence mechanism
20637659	*K33-linked polyubiquitination of TCR-ζ...*	In vitro	Molecular substrate (TCR-ζ)
20624942	*Cbl-b(-/-) T cells resistant to Tregs...*	Mouse	Independent confirmation of Treg resistance
10646608	*Negative regulation of lymphocyte activation... Cbl-b*	Mouse	Knockout multisystem autoimmunity; Vav1/CD28
10646609	*Cbl-b regulates the CD28 dependence of T-cell activation*	Mouse	EAE susceptibility; MS relevance
25815272	*Modulation of Immune Cell Functions by the E3 Ligase Cbl-b*	Review	Multi-lineage expression; rheostat framing
24553136	*Cbl-b and TAM receptors regulate cancer metastasis via NK cells*	Mouse	NK substrates; therapeutic inverse
19815501	*Cbl-b monoubiquitination of CARMA1 in NKT anergy*	Mouse/in vitro	CARMA1/NF-κB node
25000980	*Deltex1 / PKCθ / Cbl-b in anergy*	In vitro	PKCθ, PLCγ1 substrates
20453840	*CBLB variants associated with MS*	Human GWAS	Common-variant autoimmunity risk
21037273	*CBLB–MS replication (Italy)*	Human	Replication
22194214	*CBLB–MS replication (Spain)*	Human	Replication
25261476	*MS-associated CBLB variant links to type I IFN*	Human/in vitro	Mechanism of risk allele

PMID	Title (abbrev.)	Evidence type	Role in this report
15897618	*Major gene for T1D in KDP rat*	Rat	Natural Cblb-mutant model
21918539	*MHC II in rat autoimmune diabetes/thyroiditis*	Rat	Organ-specific autoimmunity
17259398	*Genetic reconstitution of T1D in rat*	Rat	Two-gene model confirmation
15629882	*CBLB variants in T1D; interaction with CTLA4*	Human	Human T1D susceptibility
15117950	*Regulation of c-Cbl ligase activity*	In vitro/ structural	Domain autoinhibition relevant to variants
10362357	*cbl-3: a new mammalian cbl family protein*	In vitro	Domain architecture (TKB/ RING)

A consistent caveat: much of the mechanistic literature uses the common gene synonym ***Cbl-b***/**Casitas B-lineage lymphoma-b** rather than the formal symbol ***CBLB***, and several SLE/MS papers describe *common-variant* or *acquired* CBL-B reduction rather than the biallelic null of ADMIO3. These support the mechanism but are not direct observations of the Mendelian disease.

Limitations and Knowledge Gaps

1. **Tiny human cohort (n = 3).** All firm human clinical claims rest on a single 2022 report. Penetrance, expressivity, full phenotypic spectrum, per-phenotype frequencies, severity distribution, natural history, prognosis and quality-of-life metrics are **unknown**.
2. **Epidemiology is unquantified.** No prevalence, incidence, carrier frequency, founder-effect, geographic or sex-ratio data exist for ADMIO3.
3. **No disease-specific treatment evidence.** Therapeutic suggestions (abatacept/CTLA4-Ig, mTOR inhibition) are mechanistic extrapolations, not trial-tested in ADMIO3.
4. **Organ-specificity extrapolation.** Thyroiditis, diabetes and alopecia are prominent in rat/mouse models; the exact organ distribution in human ADMIO3 is not yet mapped.

5. **Human NK/mast-cell contribution.** The mast-cell and NK/TAM branches are demonstrated in mice; their precise contribution to human clinical disease is inferred.
6. **Nomenclature hazard.** The serial-number disease name does not encode the gene; literature mixes *CBLB*, *Cbl-b*, and common-variant studies — care is needed to avoid conflating hypomorphic-risk-allele biology with the biallelic-null Mendelian disease.

Proposed Follow-up Experiments / Actions

1. **Expand the patient registry** via GeneMatcher/collaborative IEI networks to define phenotype frequencies, penetrance, natural history and genotype–phenotype correlations across the three index-allele classes (null vs. RING vs. TKB hypomorph).
2. **Deep immunophenotyping** of patients: standardized Treg-suppression assays, TCR-signaling thresholds, FcεRI-driven mast-cell/basophil activation tests, NK-cell TAM-axis function — to confirm which mouse-derived branches operate in humans.
3. **Variant functional characterization** in patient iPSC-derived or primary T cells with isogenic correction, quantifying E3 ligase activity and substrate (TCR-ζ, Vav1, CARMA1) ubiquitination per allele.
4. **Therapeutic proof-of-concept**: test abatacept/CTLA4-Ig and mTOR inhibition in the *CblbH257L* knock-in mouse for autoimmunity and anaphylaxis endpoints, as the preclinical bridge to compassionate-use human trials.
5. **Prognostic biomarker discovery**: longitudinal autoantibody, cytokine (IL-2) and Treg-resistance profiling to predict organ-specific flares.
6. **Carrier/population screening analysis** of gnomAD for biallelic-compatible *CBLB* LoF burden to estimate disease frequency and inform counseling.

Consensus Answer

Autoimmune disease, multisystem, infantile-onset, 3 (ADMIO3; OMIM 620430; MONDO:0957388) is a rare autosomal-recessive inborn error of immunity caused by **biallelic loss-of-function variants in *CBLB***, encoding the E3 ubiquitin ligase CBL-B — the central negative checkpoint that ubiquitinates proximal antigen-receptor signaling components to raise lymphocyte activation thresholds and enforce peripheral tolerance. Loss of CBL-B

catalytic activity lowers the activation threshold, uncouples T-cell activation from CD28 costimulation, impairs anergy, and renders effector T cells resistant to Treg suppression while hyperactivating B, NK and mast cells, producing infantile-onset multisystem autoimmunity (including allergic features such as chronic urticaria); the mechanism is corroborated by *Cblb*-knockout/knock-in mice and the naturally *Cblb*-mutant Komeda diabetes-prone rat.
